End Term-1 Exam 2025-2026 Odd Semester
Department of Computer Applications
National Institute of Technology, Kurukshetra

Roll Number _____

Course Code: CSIC 221 Subject: Machine Learning & Data Analysis Pages:16, Questions: 56	27 November 2025 09:30-12:30 Max Marks: 50 Duration: 3 Hours

Note:

- Write your Roll No. and Name on both the Question Paper and Answer Script.
- Attempt all questions in the Answer Script only — no extra copy/pages will be given.
- For MCQs, write the option letter and full answer text/option text.
- For partial/doubtful answers, show the formula used and the final value.
- If information is missing, state your assumptions clearly.
- Return both the Question Paper and Answer Script to the invigilator.
- Scientific calculators allowed; no exchange of stationery or calculators.
- All the best!

Some values that might be required for solving the questions are listed here-

- $Z_{0.05} = 1.645$
- $\chi^2_{0.05} = 5.991$ with appropriate degree of freedom

Part A

Q1 [0.5 M] You need a rank-k ($k \ll 1000$) approximation of a 1000×1000 dense matrix A for dimensionality reduction. The matrix is not guaranteed to be symmetric and numerical stability is important. Which method is most suitable?

(A) LU Decomposition
(B) Standard Eigenvalue Decomposition
(C) Truncated Singular Value Decomposition
(D) QR Decomposition

Q2 [0.5 M] Which of the following best describes the role of the learning rate (α) in the Gradient Descent algorithm?

(A) It determines the batch size for the gradient calculation.
(B) It controls the magnitude of the step taken in the direction opposite to the gradient.
(C) It sets the number of training epochs before the algorithm stops.
(D) It defines the total number of iterations before convergence.

Q3 [0.5 M] Machine learning tasks that involve predicting a continuous outcome (like a home's price) based on past data fall under the category of _____ learning.

(A) Regression
(B) Classification
(C) Reinforcement Learning
(D) Clustering

Q4 [0.5 M] Too many irrelevant features can lead to the problem known as the _____.

(A) Gradient Explosion
(B) Local Minimum Trap
(C) Bias-Variance Tradeoff
(D) Curse of Dimensionality

Q5 [0.5 M] A hospital is using a Naïve Bayes classifier to predict whether a patient has Disease (D) or No Disease (ND) based on symptoms. Prior probabilities: P(D) = 0.25, P(ND) = 0.75. A new patient reports two symptoms: Fever (x_1) and Headache (x_2). The conditional probabilities are:

Symptom	P(x | D)	P(x | ND)
Fever (x_1)	0.8	0.3
Headache (x_2)	0.6	0.4

Which class does the model predict?

(A) Disease (D)
(B) Not enough information
(C) No Disease (ND)
(D) Tie

Q6 [0.5 M] A square has a vertex at P(3,0). It is rotated 90° counterclockwise about the origin. What are the new coordinates of P?

(A) (0,3)
(B) (−3,0)
(C) (3,0)
(D) (0,−3)

Q7 [0.5 M] Which test is used to compare the means of two small samples with unknown population variances?

(A) z-test
(B) t-test
(C) F-test
(D) Chi-square test

Q8 [0.5 M] In the context of classification, Information Gain is often used as a feature selection metric because a feature with high information gain is expected to significantly reduce the dataset's overall _____.

(A) Variance
(B) Standard Deviation
(C) Entropy
(D) Learning Rate

Q9 [0.5 M] A matrix A has an eigenvector v corresponding to eigenvalue $\lambda = 6$. What is the result of applying A^4 to v?
$A^4 v$ = ?

(A) $6^4 v$
(B) 6v
(C) 24
(D) $4^6 v$

Q10 [0.5 M] In K-means clustering, the term centroid refers to the:
(A) Radius used to define the boundary of a cluster.
(B) Farthest data point from the center of a cluster.
(C) Geometric center (mean) of all the data points assigned to a cluster.
(D) Initial data point randomly selected to start the process.

Q11 [0.5 M] A data scientist performs PCA on a dataset and observes that the first principal component explains 85% of the total variance. The scientist decides to retain only this component for downstream analysis and reconstruction. What is the main practical drawback of using only the first principal component for dimensionality reduction?
(A) PCA will automatically preserve all features regardless of compression.
(B) The model becomes more computationally expensive than using all original features.
(C) The reconstructed data will perfectly match the original dataset.
(D) There may be information loss and possible reconstruction error.

Q12 [0.5 M] A data analyst is working with a geospatial dataset where clusters are irregularly shaped and may contain significant noise (outliers). Which clustering algorithm is generally the most suitable for this scenario?
(A) K-means Clustering
(B) DBSCAN Clustering
(C) Gaussian Mixture Model Clustering
(D) Hierarchical Agglomerative Clustering (using Ward's method)

Q13 [0.5 M] Consider a K-means clustering result where the Sum of Squared Errors (SSE) is very high. Which action related to the parameter k (the number of clusters) would be the most likely corrective step to decrease the SSE?
(A) Keeping *k* the same and changing the distance metric.
(B) Increasing the value of *k*.
(C) Keeping *k* the same and using a different initialization method.
(D) Decreasing the value of *k*.

Q14 [0.5 M] The Null Space of a matrix A is defined as:
(A) The set of all vectors x such that A x = 0
(B) The set of all vectors x such that A^2 x = 0
(C) The set of all vectors x such that A x = 1
(D) The set of all vectors x such that A^T x = 0

Q15 [0.5 M] Given A = $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$
find the dimension of the Null Space of A.

(A) 1
(B) 0
(C) 2
(D) 3

Q16 [0.5 M] Naïve Bayes is particularly efficient for high-dimensional data because each feature's conditional probability can be estimated independently, thereby reducing the impact of the _____.

(A) Model Underfitting
(B) Linear Dependence of Features
(C) Gradient Vanishing
(D) Curse of Dimensionality

Q17 [0.5 M] Consider a linear regression model where $y_i = \beta x_i + \varepsilon_i$, and ε_i are independent and identically distributed as $N(0, \sigma^2)$. Given the data $\{(x_1, y_1), (x_2, y_2)\} = \{(1, 2), (2, 4)\}$, and assuming σ^2 is known, what is the MLE of β?

(A) 3
(B) 2.5
(C) 2
(D) 1

Q18 [0.5 M] Consider the matrix multiplication $\mathbf{C}=\mathbf{AB}$, where $\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×4. If you want to compute the i-th column of $\mathbf{C}$ without calculating the entire product, you would express it as a linear combination of the columns of _____ weighted by the entries of the i-th column of _____.

(A) Columns of B, weighted by entries of the i-th column of A
(B) Columns of A, weighted by entries of the i-th column of B
(C) Rows of A, weighted by entries of the i-th row of B
(D) Rows of B, weighted by entries of the i-th row of A

Q19 [0.5 M] The primary mechanism by which both feature selection and feature extraction mitigate the curse of dimensionality is by reducing the effective dimensionality of the data space.

(A) False
(B) True

Q20 [0.5 M] For a perfectly symmetric normal distribution, the mean would be equal to which of the following measures?

A. Median
B. Mode
C. Second quartile
D. Third decile
E. 50th percentile

Choose the most appropriate answer from the options given below:

(A) A, B, C, and E only
(B) B, C, and D only
(C) A, B, C, D, and E
(D) A, B, and C only

Answer Q21– Q23 based on the following text:
Solve the following questions based on this matrix:

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

Q21 [1 M] The right singular vectors are obtained from which matrix?

(A) $\begin{bmatrix} 8 & -9 \\ -9 & 8 \end{bmatrix}$

(B) $\begin{bmatrix} 9 & -8 \\ -8 & 9 \end{bmatrix}$

(C) $\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$

(D) $\begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$

Q22 [1 M] The left singular vectors are obtained from which matrix?

(A) $\begin{bmatrix} 2 & -4 & 4 \\ -4 & 8 & -8 \\ 4 & -8 & 8 \end{bmatrix}$

(B) $\begin{bmatrix} 9 & -9 & 9 \\ -9 & 9 & -9 \\ 9 & -9 & 9 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & -3 & 3 \\ -3 & 6 & -6 \\ 3 & -6 & 6 \end{bmatrix}$

(D) $\begin{bmatrix} 6 & -3 & 3 \\ -3 & 6 & -6 \\ 3 & -6 & 6 \end{bmatrix}$

Q23 [1 M] Given a matrix A, its singular values are found to be approximately:

(A) $sqrt(9)$,0

(B) $2/sqrt(3)$,0

(C) $3.sqrt(2)$,0

(D) $2.sqrt(2)$,0

Answer Q24– Q25 based on the following text:
Consider the given set of vectors. (Hint: To optimize the time to solve the questions in the group, read both the questions of the group before solving any.) [5]

$$v_1^t = [1\ 1\ 1\ 0]; v_2^t = [1\ 2\ 3\ 1]; v_3^t = [-1\ 0\ 1\ -1]; v_4^t = [0\ 1\ 2\ -1]$$

Q24 [3 M] Find the orthonormal basis (consider each column to be a vector)

(A) $\frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 0 & -1 & -1 & 1 \end{pmatrix}$

(B) $\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

(D) $\frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & \frac{-1}{\sqrt{2}} & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & \frac{-2}{\sqrt{2}} & 0 \end{pmatrix}$

Q25 [2 M] State the cardinality of the set of linearly independent vectors that can make all four vectors.
(A) 3 (B) 5 (C) 4 (D) 2

Answer Q26– Q28 based on the following text:
Ervin bowled 7 games last weekend. His scores (in points) were:
155, 165, 138, 172, 127, 193, 142
Use the data to answer the following questions:
Q26 [0.5 M] What is the population variance of Ervin's bowling scores?

(A) 511.33
(B) 438.29
(C) 22.61
(D) 20.94

Q27 [0.5 M] What is the sample standard deviation of Ervin's bowling scores?

(A) 511.33
(B) 22.61
(C) 438.48
(D) 20.94

Q28 [0.5 M] What is the correct mathematical relationship between variance and standard deviation?
(A) The variance is the square root of the standard deviation.
(B) The variance is the difference between the standard deviation and the average.
(C) The standard deviation is the square root of the variance.
(D) The standard deviation is the square of the difference between the variance and the average.

Answer Q29– Q32 based on the following text:
Using the following actual and predicted class labels:[2+0.5+0.5+0.5]

Actual Class	1	1	0	0	0	1	0
Predicted Class	1	1	0	0	1	1	0

Q29 [0.5*4 M] Fill in the missing values in the confusion matrix below:

	Predicted: 1	Predicted: 0
Actual: 1	TP = ___	FN = ___
Actual: 0	FP = ___	TN = ___

Q30 [0.5 M] The precision for Class 1 is closest to:

(A) 0.60
(B) 0.50
(C) 0.75
(D) 1.00

Q31 [0.5 M] The recall for Class 1 is:

(A) 0.75
(B) 1.00
(C) 0.67
(D) 0.50

Q32 [0.5 M] The F1-Score for Class 1 based on the given confusion matrix is approximately:

(A) 0.85
(B) 0.50
(C) 0.67
(D) 1.00

Answer Q33– Q37 based on the following text:
Consider the following 6 points plotted in a 2D space (Euclidean distance assumed):

Point	Coordinates
P1	(1,1)
P2	(1,2)
P3	(2,2)
P4	(8,9)
P5	(9,9)
P6	(25,30)

DBSCAN parameters: eps = 1.2; minPts = 3
Q33 [0.5 M] If eps is increased to 2.5 (minPts remains 3), what becomes true?
(A) No changes in cluster assignments occur
(B) All points become one large cluster
(C) P4 and P5 now form a valid cluster
(D) P6 joins the existing cluster

Q34 [0.5 M] Which points form the first valid DBSCAN cluster?

(A) {P4, P5}
(B) {P1, P2}
(C) {P1, P2, P3}
(D) No cluster is formed

Q35 [0.5 M] How many neighbors does P1 have within eps?

(A) 0
(B) 1

(C) 3 (D) 2

Q36 [0.5 M] Which points qualify as core points under current settings?

(A) P1, P2, and P3
(B) All points except P6
(C) Only P2
(D) P2 and P3

Q37 [0.5 M] Under the current eps and minPts values, which points will be labelled as noise?

(A) P4, P5, and P6
(B) None of the points
(C) P1 and P6
(D) Only P6

Part B

Q38 [2 M] For every situation in Group B, choose all possible outcomes from Group A.

Group A	Group B	Answer
1. Unique solution	No free variables	
2. Inconsistent Equations	b. Coefficient matrix has lower rank than augmented matrix	
3. Infinitely many solutions	c. Coefficient matrix and augmented matrix has same rank	
	d. Coefficient matrix's Column vectors are linearly independent	

Q39 [1 M] To impute the missing value (TotalSMS for Customer 4) using KNN Imputation with k = 2, we identify the two nearest neighbours based on the other available features: [1]
Relevant numeric values required:

Customer	TotalSMS	TotalData(GB)	BillAmount
1	120	2.5	799
2	80	1.8	620
3	140	2.9	950
4	NaN	2.6	885

Q40 [11*0.5 M] Identify the type of data analysis problem (Classification, Regression, Clustering, Dimensionality Reduction, Similarity Measure) that the following techniques/algorithms are used for.

S.No	Data Analysis Algorithm / Technique	Data Analysis Problem Type	Marks
1	K-Means	_____	½
2	Logistic Regression	_____	½
3	PCA (Principal Component Analysis)	_____	½
4	DBSCAN	_____	½
5	Correlation coefficients	_____	½
6	Hierarchical Clustering	_____	½
7	Knn	_____	½
8	Pearson Correlation	_____	½
9	Decision Tree	_____	½
10	Naive bayes	_____	½
11	Linear Regression	_____	½

Answer Q41– Q45 based on the following text:
A machine learning model is being trained using Gradient Descent to minimize the following loss function:
$L(w) = (w-5)^2$
The initial weight is:
$w_0 = 0$
The learning rate is:
α=0.2
Perform two iterations of Gradient Descent and answer the following:
Q41 [1 M] Compute the gradient value at w_0.

Q42 [1 M] Compute the updated weight after the second iteration.

w_2 =_____

Q43 [1 M] After both iterations, the updated weight is _____:

(A) Not changing (stuck)
(B) Moving away from the minimum
(C) Oscillating
(D) Moving toward the minimum

Q44 [1 M] If the stopping criteria is:
$|w_{t+1} - w_t| < 0.05$
Does Gradient Descent stop after iteration 2?

(A) No
(B) Yes

Q45 [1 M] Compute the updated weight after the first iteration.
w_1 =_____

Answer Q46– Q50 based on the following text:
A survey was conducted among 120 students to investigate whether their branch of study influences their preference for machine learning algorithms. The observed frequencies are summarized in the contingency table below: [5]

Algorithm Preference	CSE	ECE	Total
Decision Tree	30	10	40
K-Means	20	25	45
Logistic Regression	10	25	35
Total	60	60	120

Using a significance level of α = 0.05, answer the following questions:
Q46 [1 M] Identify the most appropriate statistical test for analyzing this dataset.

Q47 [1 M] Determine the degrees of freedom for this test.

Q48 [2 M] Compare the calculated test statistic with the critical value (OR interpret the p-value), and state whether the variables are independent or dependent.

Q49 [1 M] Calculate the expected frequency for the cell corresponding to Decision Tree – CSE.

Q50 [1 M] Clearly state the null hypothesis (H_0) and the alternative hypothesis (H_1) for this test.

Answer Q51– Q53 based on the following text:
Six points represent customer behaviour in a 2-dimensional feature space:

Sample	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Using Agglomerative Hierarchical Clustering with Single-Link (minimum distance) and Euclidean distance, perform the clustering and answer the following:
Q51 [1 M] Compute the updated weighted matrix after 1st iteration.
Matrix: _____

Q52 [2 M] Draw the resulting Single-Link(final) Dendrogram based on the merge order.

Q53 [2 M] Compute the pairwise distance between the closest pair of points after 1st iteration.
Closest pair: _____
Distance: _____

Answer Q54– Q56 based on the following text:
The speed of wind in bad weather events is assumed to follow a normal distribution with a population standard deviation of $\sigma = 8$ km/h. In one year, a random sample of 44 bad weather events gives a sample mean wind speed of 5.4 km/h and a sample standard deviation of 7.3023 km/h. Using $\alpha = 0.05$, test whether the true mean wind speed in bad weather events is significantly greater than an expected speed of 3 km/h.
State the:
Q54 [1 M] Test statistic:

Q55 [1 M] Null and alternative hypotheses:

Q56 [2 M] Critical value / p-value and Decision:

ANSWER KEY - SET 1

Q1 [0.5 M] → Ans: Truncated Singular Value Decomposition (Truncated SVD) (Note: did not find exact OPT match after shuffle)
Q2 [0.5 M] → Ans: It controls the magnitude of the step taken in the direction opposite to the gradient.
Q3 [0.5 M] → Ans: Regression
Q4 [0.5 M] → Ans: Curse of Dimensionality
Q5 [0.5 M] → Ans: Disease (D)
Q6 [0.5 M] → Ans: (0,3)
Q7 [0.5 M] → Ans: t-test
Q8 [0.5 M] → Ans: Entropy
Q9 [0.5 M] → Ans: $6^4 v$
Q10 [0.5 M] → Ans: Geometric center (mean) of all the data points assigned to a cluster.
Q11 [0.5 M] → Ans: There may be information loss and possible reconstruction error.
Q12 [0.5 M] → Ans: DBSCAN Clustering
Q13 [0.5 M] → Ans: Increasing the value of k. (Note: did not find exact OPT match after shuffle)
Q14 [0.5 M] → Ans: The set of all vectors x such that A x = 0
Q15 [0.5 M] → Ans: 1
Q16 [0.5 M] → Ans: Curse of Dimensionality
Q17 [0.5 M] → Ans: 2
Q18 [0.5 M] → Ans: Columns of A, weighted by entries of the i-th column of B
Q19 [0.5 M] → Ans: True
Q20 [0.5 M] → Ans: OPT: A, B, C, and E only (Note: did not find exact OPT match after shuffle)
Q21 [1 M] → Ans: $\begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$
Q22 [1 M] → Ans: $\begin{bmatrix} 2 & -4 & 4 \\ -4 & 8 & -8 \\ 4 & -8 & 8 \end{bmatrix}$
Q23 [1 M] → Ans: $3.sqrt(2)$,0
Q24 [3 M] → Ans: $\frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & \frac{-1}{\sqrt{2}} & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & \frac{-2}{\sqrt{2}} & 0 \end{pmatrix}$
Q25 [2 M] → Ans: 3
Q26 [0.5 M] → Ans: 438.29
Q27 [0.5 M] → Ans: 22.61
Q28 [0.5 M] → Ans: The standard deviation is the square root of the variance (Note: did not find exact OPT match after shuffle)
Q29 [0.5*4 M] →

Pred 1 Pred 0

Actual 1	TP = 3	FN = 0
Actual 0	FP = 1	TN = 3

Q30 [0.5 M] → Ans: 0.75
Q31 [0.5 M] → Ans: 1.00
Q32 [0.5 M] → Ans: 0.85
Q33 [0.5 M] → Ans: No changes in cluster assignments occur
Q34 [0.5 M] → Ans: {P1, P2, P3}. (Note: did not find exact OPT match after shuffle)
Q35 [0.5 M] → Ans: 1
Q36 [0.5 M] → Ans: Only P2
Q37 [0.5 M] → Ans: P4, P5, and P6

Part B

Q38 [2 M] → a: (1-2), b: 2, c: (1-3), d: (1-2)
Q39 [1 M] → NaN: 130
Q40 [11*0.5 M] →
Clustering;
Classification;
Dimensionality Reduction;
Clustering;
Dimensionality reduction;
Clustering;
Classification/Regression;
Feature Selection / Dimensionality Reduction/Similarity measure;
Classification;
Classification;
regression
Q41 [1 M] → Gradient: -10
Q42 [1 M] → w_2: 3.2
Q43 [1 M] → Ans: Moving toward the minimum
Q44 [1 M] → Ans: No
Q45 [1 M] → w_1: 2
Q46 [1 M] → Test: Chi-Square Test of Independence
Q47 [1 M] → Degrees of freedom: 2
Q48 [2 M] → p-value: 5.991, Decision: Reject H_0
Q49 [1 M] → Expected Frequency: 20
Q50 [1 M] → Null Hypothesis (H_0): Algorithm preference is independent of the student's branch (CSE/ECE), Alternative Hypothesis (H_1): Algorithm preference depends on the student's branch (they are associated).
Q51 [1 M] → Matrix:
Q52 [2 M] → Dendrogram:
Q53 [2 M] → Closest pair: P3 and P6, Distance: 0.10

Q54 [1 M] → Test value: 1.99
Q55 [1 M] → H_0 (null): $\mu = 3$, H_1 (alternative): $\mu > 3$ (right-tailed test)
Q56 [2 M] → p-value: 0.0233, Decision: reject H_0.